

Erdős Problem #688

EP-688

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Define ϵ_n to be maximal such that there exists some choice of congruence class a_p for all primes $n^{\epsilon_n} < p \leq n$ such that every integer in $[1, n]$ satisfies at least one of the congruences $\equiv a_p \pmod{p}$.

Estimate ϵ_n - in particular is it true that $\epsilon_n = o(1)$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>